

KNMTG: A Korean Nationwide Multimodal Transit Graph for Time-Dependent Midpoint Search

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Abstract

We present KNMTG (Korean Nationwide Multimodal Transit Graph), a time-dependent weighted graph unifying five public transportation modes across South Korea: urban subway, KTX/SRT high-speed rail, regional rail (Mugunghwa/Saemaeul/ITX/GTX-A), intercity express bus, and bus rapid transit (BRT). The graph comprises 932 nodes—spanning 586 urban metro stations, 47 KTX/SRT/GTX-A terminals, 254 regional rail stops, 42 intercity bus terminals, and 3 BRT hubs—connected by 1,458 directed edges with empirically calibrated travel times and transfer penalties. Building on this graph, we develop a *symmetric midpoint finder* that identifies optimal meeting stations minimizing the maximum travel time across multiple parties. A web-based demonstration supporting multi-origin queries, share links, favorites, and taxi-access mode is publicly accessible. Validation against 12 real-world origin–destination pairs spanning all five modes shows a mean absolute error of 3.8 minutes relative to published timetables and commercial navigation apps. The full dataset and code are released under CC-BY 4.0 and MIT licenses, respectively, at <https://github.com/yjkim0123/knmtg> to support reproducible transit research in Korea.

Keywords: multimodal transit graph; Korean public transportation; time-dependent routing; meeting point problem; dataset

1 Introduction

Finding a fair meeting point for geographically dispersed individuals is a recurring problem in transportation research (Yan et al., 2011; Ahmadi and Nascimento, 2018). While road-network solutions are well-studied, transit networks introduce structural complexity: heterogeneous modes, discrete timetables, transfer penalties, and nonlinear fare structures. In South Korea, the challenge is amplified by the coexistence of one of the world’s densest urban subway systems (Seoul metro: 23 lines, 750+ stations), a nationwide KTX high-speed

rail network, and an extensive intercity bus grid—yet no unified, publicly available graph integrating all three exists.

Existing Korean transit datasets are either (1) limited to a single city’s subway (Lee et al., 2008; Kim and Jung, 2019), (2) restricted to one mode (bus or rail) (Lee and Jung, 2012), or (3) commercially proprietary (Naver/Kakao routing APIs). Open pan-modal graph datasets for other countries (Kujala et al., 2018) stop at GTFS boundaries, missing the intercity infrastructure that dominates long-distance Korean trips.

This paper makes three contributions:

1. **KNMTG dataset:** A unified, open, time-dependent transit graph for South Korea covering five transportation modes, with empirically calibrated edge weights and transfer penalty parameters.
2. **Routing algorithm:** A transfer-penalty-aware Dijkstra implementation that correctly models KTX boarding overhead, headway-based waiting, and multimodal path composition.
3. **Midpoint application:** A web-based symmetric meeting-point finder evaluated on real-world cases demonstrating the graph’s practical utility.

2 Related Work

2.1 Transit Graph Datasets

Derrible and Kennedy Derrible and Kennedy (2009) analyzed the topological properties of 19 world subway systems (including Seoul) using graph metrics, establishing a comparative baseline. Lee et al. (2008) examined Seoul subway passenger flows as a complex network. For intercity connectivity, Lee and Jung (2012) constructed and analyzed Korea’s intercity express bus network as a weighted graph. Kujala et al. (2018) released GTFS-derived graph datasets for 25 cities in standardized format; our work follows this open-dataset spirit while extending coverage to non-GTFS modes (KTX timetables, BRT, intercity bus).

2.2 Multimodal Routing

Bast et al. (2016) provide a comprehensive survey of route planning algorithms for large transportation networks, including transfer graph abstractions for heterogeneous modes. Delling et al. (2015) introduced RAPTOR for round-based schedule-aware routing; Witt (2015) developed trip-based routing with Pareto-optimal queries; Baum et al. (2021) proposed ULTRA for unlimited multi-modal transfers without precomputation overhead. Our implementation adopts a Dijkstra-based approach with explicit transfer penalties rather than RAPTOR, trading worst-case optimality for simplicity of multi-origin aggregation.

2.3 Meeting Point Problems

Papadias et al. (2004) introduced the group nearest neighbor problem on road networks minimizing aggregate travel cost. Yan et al. (2011) extended this to min-max fairness for-

mulations and proposed efficient Dijkstra-based algorithms. Ahmadi and Nascimento (2018) were the first to address meeting-point queries for public transit users, accounting for schedules and transfers. Our application implements the min-max fairness criterion (minimize the maximum individual travel time) with an optional balanced-variance scoring function.

3 Graph Construction

3.1 Node Types and Sources

KNMTG nodes represent physical transit stops or terminals of five types (Table 1). Subway station coordinates and line membership were extracted from official GTFS feeds supplemented by manual curation for newly opened lines. KTX/SRT/GTX-A terminals and transfer times were compiled from KORAIL and SR timetable releases. Regional rail stops (Mugunghwa, Saemaeul, ITX-Maeum) were derived from published route alignments. Intercity bus terminals and BRT hubs were sourced from the Ministry of Land, Infrastructure and Transport’s TAGO bus route API.

Table 1: Node composition of KNMTG v1.1

Type	Description	Count
Subway	Urban metro stations (Seoul, Busan, Daegu, Gwangju, Daejeon, Incheon)	586
KTX/SRT/GTX-A	High-speed rail and express metro terminals	47
Regional Rail	Mugunghwa/Saemaeul/ITX-Maeum stops	254
Bus Terminal	Intercity express bus terminals	42
BRT Hub	Bus rapid transit boarding points	3
Total		932

3.2 Edge Construction

Subway edges are constructed from line segment lists with empirically measured per-stop travel times derived from GTFS schedules. For regional subway systems (Busan, Daegu, Gwangju, Daejeon), average inter-station times were inferred from published route lengths and operating speeds.

KTX/SRT edges use segment-level travel times published in official timetables. A transfer penalty $p_{KTX} = 20$ minutes is added to every KTX boarding event, reflecting ticketing, platform access, and minimum connection time. This penalty was calibrated so that bus alternatives to nearby subway stations are not spuriously preferred by the routing algorithm (Section 5).

Intercity bus edges connect 151 terminal pairs with travel times from operator timetables. A boarding penalty of 15 minutes models ticket purchase and pre-departure waiting.

City bus edges were obtained by calling the TAGO `BusRouteInfoInquireService` API for 49 cities, filtering routes whose endpoints contained hub keywords (*terminal/station/airport*), and matching bus stop coordinates within 1 km of existing graph nodes. This yielded 58 new inter-node city bus edges after deduplication.

BRT edges connect 21 BRT hubs via published route segments and link each hub to the nearest subway or KTX station.

3.3 Transfer Penalty Design

A naive implementation treats KTX-to-subway transfers as zero-cost adjacency, allowing the router to exploit bus shortcuts and circumvent the intended KTX penalty. We implement a systematic bypass-detection check: for each KTX node k , bus-mediated 2-hop paths $k \xrightarrow{t_1} m \xrightarrow{t_2} s$ to subway nodes s are enumerated; edges where $t_1 + t_2 < p_{KTX} + p_{base}$ are removed. Two such bypass edges were identified and removed during construction (Section 7).

4 Graph Properties

Table 2 summarizes the structural properties of KNMTG. The degree distribution is right-skewed with median degree 2 (chain- like subway segments) and a long tail reaching degree 41 (Express Bus Terminal / Gosok-terminal, a major multimodal hub in Seoul).

Table 2: Structural statistics of KNMTG v1.1

Property	Value
Nodes	932
Edges (undirected)	1,458
Average degree	3.13
Median degree	2
Maximum degree	41 (, Seoul)
Isolated nodes	0

Hub nodes by degree. The five highest-degree nodes are (41), (22), (22), (21), and _KTX (20). 's exceptional degree reflects its role as the primary Seoul intercity bus terminus, connected to numerous out-of-Seoul intercity routes.

Betweenness centrality. Approximate betweenness centrality (sampled from 300 source nodes) identifies / _SRT (BC=67,603), _KTX (60,906), and (60,171) as the highest-betweenness nodes—consistent with their role as national transit bottlenecks on the Gyeongbu KTX corridor. The prominence of _SRT (BC=47,805) reflects the SRT's role in connecting the southern metropolitan area with the national network.

5 Routing Algorithm

Given a set of origins $O = \{o_1, \dots, o_n\}$, each mapped to its nearest transit station, the algorithm performs n single-source Dijkstra traversals to compute travel-time maps $d_i : V \rightarrow \mathbb{R}_{\geq 0}$.

For each candidate meeting station v , a composite score is computed:

$$\text{score}(v) = w_{\max} \cdot \max_i(d_i(v)) + w_{\text{sum}} \cdot \sum_i d_i(v) + w_{\text{std}} \cdot \sigma(d_i(v)) - w_{\text{pr}} \cdot \text{PageRank}(v) \quad (1)$$

where weights w are tuned per strategy (balanced / minimize-max / minimize-sum). PageRank over the transit graph is used as a proxy for node accessibility, penalizing remote but equidistant candidates in favor of well-connected meeting hubs.

Geographic pre-filter. To reduce the candidate set from all 924 nodes, stations outside the convex hull of origin coordinates (expanded by 20%) are excluded. This reduces query time from $O(|V|^2 \log |V|)$ to $O(n \cdot |V| \log |V|)$ in practice.

Complexity. Each Dijkstra runs in $O((|V| + |E|) \log |V|)$. For $n = 2$ origins and $|V| = 924$, $|E| = 1450$, a query completes in under 50 ms in a Python reference implementation and under 5 ms in the browser JavaScript port.

6 Application: Midpoint Finder

The KNMTG midpoint finder is deployed as a web application at <https://midpoint-kr.fly.dev>, accessible from desktop and mobile browsers. Key features include:

- **Multi-origin support:** up to 4 simultaneous origins with free-text geocoding (Kakao local search API).
- **Strategy selection:** balanced (default), minimize-max, or minimize-sum scoring.
- **Time-of-day query:** departure hour selection for headway-aware routing.
- **Share links:** URL parameter encoding of all inputs for persistent sharing.
- **Favorites:** localStorage-based bookmarking of up to 12 stations.
- **Taxi access mode:** optional taxi travel time and fare estimate for last-mile access to/from transit stations.

6.1 Case Studies

CS1: Seoul (Gangnam) $\leftrightarrow$ Busan (Seomyeon). The optimal meeting point is Daejeon Station (subway), reachable in 82 min from Gangnam and 99 min from Seomyeon, placing the equidistant point near the geographic midpoint of the 325 km Seoul–Busan KTX corridor.

CS2: Incheon (Bupyeong) $\leftrightarrow$ Daegu (Dong-Daegu). The algorithm recommends Asan Station (max 82 min), reachable via the Seoul metropolitan network and the KTX corridor, rather than a geographically central point, demonstrating that transit topology deviates significantly from Euclidean midpoints.

CS3: Three-party meeting (Suwon + Uijeongbu + Gangnam). The optimal point is Seongnam_GTX (max 39 min), enabled by GTX-A connectivity from all three origins—illustrating how the GTX-A line, integrated for the first time in a Korean midpoint dataset, changes intra-metropolitan accessibility.

7 Validation

We compared KNMTG routing outputs against published timetables and real-world transit apps (Naver Maps / Kakao Maps) for 12 origin– destination pairs spanning all five transportation modes (Table 3).

Table 3: Validation of KNMTG travel times against reference values

Route	Mode	KNMTG (min)	Reference (min)	Error
_KTX $\rightarrow$ _KTX	KTX	141	~ 150	−9
_KTX $\rightarrow$ _KTX	KTX	56	~ 48	+8
_KTX $\rightarrow$ _KTX	KTX	110	$\sim 88\text{--}110$	± 0
_KTX $\rightarrow$ _KTX	KTX	101	~ 95	+6
$\rightarrow$	Saemaul	22	~ 22	0
$\rightarrow$ ()	Metro	42	~ 41	+1
$\rightarrow$	Metro	39	$\sim 30\text{--}40$	± 0
$\rightarrow$	Metro	14	$\sim 5\text{--}7$	+7
_KTX $\rightarrow$ ()	Transfer	10	~ 10	0
$\rightarrow$ ()	Bus link	15	~ 15	0
$\rightarrow$	City bus	5	~ 5	0
$\rightarrow$	BRT	25	~ 30	−5
Mean Absolute Error				3.8 min
RMSE				5.5 min

The mean absolute error is 3.8 minutes across validated routes. Larger errors occur for metro routes where per-stop times are approximated (− error: 7 min) and for KTX routes where we use average rather than fastest-service times. Two known limitations are acknowledged: (1) is not yet mapped as a subway node, leaving the Busan metro 1-line partially disconnected from _KTX; (2) (Korail) lacks a direct transfer link to Daegu urban metro, requiring future addition of a cross-mode transfer edge.

8 Limitations and Future Work

- **Static graph:** KNMTG does not incorporate real-time disruptions, platform changes, or seasonal timetable revisions. Integration with GTFS-RT feeds is a natural extension.
- **Missing KTX seat availability:** The graph models travel time but not seat supply; during peak holidays, preferred trains may be sold out.
- **Walk-time approximation:** Origin-to-station walking times are estimated from crow-fly distance; pedestrian routing (OpenStreetMap) would improve accuracy.
- **Fare model:** Ticket prices are approximated by distance bracket. Integration with the official KORAIL and intercity bus fare APIs would enable exact fare queries.
- **City bus coverage:** The current TAGO city bus scraper covers 49 cities; rural areas with sparse API data remain underconnected.
- **Mobile app:** The current deployment is a mobile- responsive web app; native iOS/Android packaging would improve usability and offline access.

9 Conclusion

We introduced KNMTG, the first publicly available, time-dependent multimodal transit graph for South Korea, integrating subway, KTX, regional rail, intercity bus, and BRT into a single weighted graph of 932 nodes and 1,458 edges. The graph is validated against real-world travel times with a mean absolute error of 3.8 minutes. Building on KNMTG, we demonstrated a symmetric midpoint-finder application with practical utility for multi-party meeting coordination. The dataset, routing code, and web application are released openly to support reproducible transit research and citizen applications.

Data Availability Statement

The KNMTG v1.1 dataset (node list, edge list, travel-time parameters) and routing source code are publicly available at <https://github.com/yjkim0123/knmtg> under CC-BY 4.0 (data) and MIT (code) licenses. The web application is accessible at <https://midpoint-kr.fly.dev>.

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